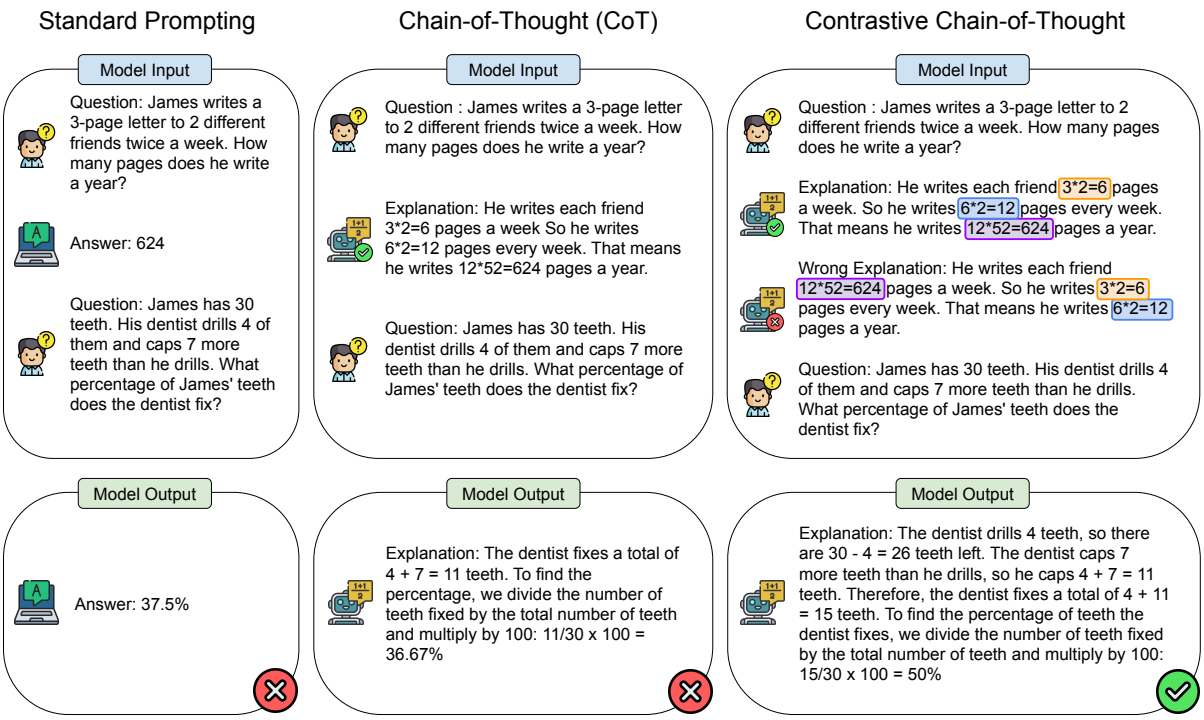

Figure 3: Overview of contrastive chain-of-thought (right), with comparison to common prompting methods.

Prompting Method	GSM8K	Bamboogle	Avg.
Standard	27.4	11.2	19.3
Chain-of-Thought	69.2	40.8	55.0
w/ Invalid Reasoning	76.0	45.6	60.8
w/ Incoherent Objects	79.6	**53.6**	**66.6**
w/ Incoherent Language	78.8	52.8	65.8
w/ Irrelevant Objects	79.8	48.8	64.3
w/ Irrelevant Language	**80.2**	49.6	64.9

Table 1: Preliminary results on the effect of contrastive demonstrations for chain of thought.

on GSM8K and Bamboogle. This suggests that language models are better able to learning step-by-step reasoning when provided with both valid and invalid rationales. Hence, we believe that contrastive demonstrations have the potential to greatly enhance language model reasoning ability.

3 Contrastive Chain of Thought

Chain-of-thought (CoT) prompting, as evidenced by prior research, has indeed elevated the reasoning capabilities of large language models (Wei et al., 2022b). However, a comprehensive understanding of this phenomenon is still lacking. Although logically sound reasoning appears to be inherently crucial for chain of thought, prior studies surprisingly reveal minimal impact when employing invalid demonstrations. To this end, based on our preliminary study in Section 2, we found that providing both valid and invalid reasoning demonstrations in a "contrastive" manner greatly improves reasoning performance. However, this approach may not generalize well to new tasks, as it requires manual construction of the invalid rationales.

Thus, we propose a general prompting method known as contrastive chain of thought, which includes automatic construction of contrastive demonstrations. Figure 3 presents an overview of our approach. Specifically, the language model is provided with the question, ground truth answer explanation and incorrect answer explanation. Compared to standard prompting, our method enables models to perform more complex reasoning by decomposing problems into intermediate steps. Compared to conventional chain-of-thought prompting, our method contrasts the valid and invalid answer explanations, guiding the model to generate more accurate reasoning chains.

Concretely, given a small set of n in-context demonstration examples $D = \{E_1, \ldots, E_{|n|}\}$, and a query Q, the goal of the model is to generate a suitable answer A. For standard prompting, the demonstration examples consist of just the question and answer, i.e., $E_j = (Q_j, A_j)$. On the other hand, chain-of-thought is a more advanced prompting method that guides the model with intermediate

Prompting Method	*Arithmetic Reasoning*					*Factual QA*	
	GSM8K	**AQuA**	**GSM-Hard**	**SVAMP**	**ASDIV**	**Bamboogle**	**StrategyQA**
Standard	27.4	29.5	11.2	69.3	75.8	12.0	59.4
CoT	69.2	53.5	33.8	67.2	70.8	40.8	55.8
Contrastive CoT	79.0 (+9.8)	57.5 (+3.9)	44.2 (+10.4)	81.6 (+14.4)	84.4 (+13.6)	56.8 (+16.0)	66.2 (+10.4)
Standard-SC	28.0	29.9	11.0	69.0	76.0	11.2	59.6
CoT-SC	71.0	55.9	34.0	71.6	74.0	40.8	57.0
Contrastive CoT-SC	86.2 (+15.2)	71.7 (+15.7)	50.0 (+16.0)	85.2 (+13.6)	89.6 (+15.6)	58.4 (+17.6)	69.6 (+12.6)

Table 2: Main evaluation results for contrastive chain-of-thought on several reasoning tasks.

Dataset	Type	|Train|	|Test|
GSM8K	Arithmetic Reasoning	4	500
AQuA	Arithmetic Reasoning	4	254
GSM-Hard	Arithmetic Reasoning	4	500
SVAMP	Arithmetic Reasoning	4	500
ASDIV	Arithmetic Reasoning	4	500
Bamboogle	Factual QA	4	125
StrategyQA	Factual QA	4	500

Table 3: Details of datasets used.

reasoning steps T. As shown in the figure above, the reasoning steps T typically consist of multiple sentences where each sentence describes one reasoning step. Hence, chain-of-thought prompting examples consist of the question, reasoning steps, and final answer, i.e., $E_j = (Q_j, T_j, A_j)$. However, the model does not know what faults to avoid in conventional chain-of-thought, which could lead to increased mistakes and error propagation. Hence, our contrastive chain of thought method provides both the correct and incorrect reasoning steps in the demonstration examples, i.e., $E_j = (Q_j, T_{j,+}, A_{j,+}, T_{j,-}, A_{j,-})$.

To obtain the correct reasoning steps T_+ for the demonstration examples, we use the annotated examples from the previous chain-of-thought works. For the incorrect reasoning steps T_-, we automatically construct it from the correct reasoning steps T_+, based on the "Incoherent Objects" category in Section 2. Concretely, we use an existing entity recognition model[5] to extract the object spans such as numbers, equations, or persons from a given chain-of-thought rationale. Consequently, we randomly shuffle the position of the objects within the rationale, thus constructing a rationale with incoherent bridging objects. Note that when testing with a new question, only the question and demonstration examples are provided to the model, and the model must generate its own reasoning steps before producing the final answer.

[5] https://spacy.io/models/en#en_core_web_trf

4 Experiments

4.1 Experimental Setup

We focus our study on two main types of reasoning tasks: arithmetic reasoning and factual question answering (QA). For arithmetic reasoning, we conduct experiments on a range of datasets including GSM8K (Cobbe et al., 2021), AQuA (Ling et al., 2017), GSM-Hard (Gao et al., 2023), SVAMP (Patel et al., 2021), and ASDIV (Miao et al., 2020). For factual QA, we include two datasets: Bamboogle (Press et al., 2023) and StrategyQA (Geva et al., 2021). To maintain a reasonable computing budget, we limit each dataset to a maximum of 500 test samples through random sampling. For datasets that contain less than 500 test samples, we instead use all available test samples. The datasets' details are included in Table 3. Regarding model and prompting details, we use the same experimental setup as for our preliminary study in Section 2.

4.2 Main Results

To assess the effectiveness of our method, we evaluate on several reasoning tasks and report the main results in Table 2. Our main findings are as follows:

Contrastive CoT demonstrates consistent improvements over conventional CoT. Contrastive CoT consistently outperforms conventional CoT across the datasets in both arithmetic and factual reasoning categories. Notably, we observe substantial gains of more than 10 points on GSM-Hard, SVAMP, ASDIV, Bamboogle and StrategyQA. Thus, the consistent and significant performance improvements demonstrate the general effectiveness of our proposed method. As contrastive chain of thought can be automatically constructed from existing rationales, the annotation cost is the same as conventional chain of thought. Hence, it

can be viewed as a general enhancement of chain of thought.

Contrastive CoT is more effective when applied with self-consistency. As self-consistency (Wang et al., 2022) is a popular decoding strategy to boost the chain-of-thought performance of large language models, we are interested to see if contrastive chain of thought can benefit similarly from self-consistency. In general, we observe that self-consistency further enhances the performance of contrastive CoT. This enhancement is particularly evident in the case of the AQuA dataset. While contrastive CoT alone results in a modest performance improvement of 4.0%, applying self-consistency amplifies this gain significantly, achieving an additional improvement of 14.2%.

5 Related Work

Large Language Models Recent developments in large language models have shown that massively scaling the size and training data of models can greatly improve generalization (Kaplan et al., 2020). Notably, large language models have been shown to generalize to new tasks when given suitable prompts and demonstrations (Brown et al., 2020). This has brought about a new paradigm of leveraging language models for tasks without the need for additional training (Liu et al., 2023). However, simply scaling language models has not been sufficient to attain good performance on challenging tasks such as arithmetic reasoning and factual question answering (Wei et al., 2022b). Hence, in this work, we focus on enhancing the reasoning ability of large language models through prompts.

Chain of Thought Chain-of-thought prompting was introduced by Wei et al. (2022b) to enhance language model reasoning by generating intermediate steps. Notably, this has inspired numerous works that build upon this direction of step-by-step reasoning. For instance, automatic chain-of-thought (Zhang et al., 2023) was proposed to address the challenges in manually annotating chain-of-thought demonstrations. On the other hand, it was shown that specific prompts such as "Let's think step-by-step" can enable language models to perform chain-of-thought in a zero-shot manner, without any demonstrations (Kojima et al., 2022). In addition, challenging problems can be decomposed into multiple sub-problems (Zhou et al., 2023), or even into code programs that can be automatically executed (Gao et al., 2023). Despite the progress in chain-of-thought on multiple fronts, we still lack a rigorous understanding of the underlying mechanism (Turpin et al., 2023; Feng et al., 2023). In this work, inspired by the findings of previous works regarding invalid demonstrations, we propose contrastive chain-of-thought to enhance language model reasoning. As contrastive chain-of-thought leverages both valid and invalid reasoning demonstrations, we believe this may encourage other researchers to fundamentally rethink the chain-of-thought process.

Learning from Negative Examples While chain-of-thought prompting typically involves only valid demonstrations, it is not clear whether invalid demonstrations can also benefit the reasoning process (Wang et al., 2023). On the other hand, learning from negative or invalid samples is not new. For instance, contrastive learning is a well-established deep learning approach that encourages models to distinguish between "positive" and "negative" samples, thus learning better representations (Khosla et al., 2020). Similarly, reinforcement learning from human feedback (RLHF) trains a reward model based on positive and negative samples of human preference data (Ouyang et al., 2022; Christiano et al., 2017). Hence, inspired by the previous approaches, we propose contrastive chain-of-thought, a general enhancement of chain-of-thought prompting, by enabling models to learn from both valid and invalid reasoning demonstrations.

6 Conclusions

In this work, we have explored the effect of leveraging invalid reasoning demonstrations for enhancing chain of thought. Through our preliminary study on different invalid chain-of-thought categories, we found that providing both valid and invalid demonstrations in a contrastive manner greatly improves reasoning ability in language models. To overcome the challenge of manually annotating invalid rationales, we propose contrastive chain of thought, a general prompting method which can automatically construct contrastive demonstrations from existing rationales. Through experiments on several reasoning tasks, we find contrastive chain of thought to be a general enhancement of chain-of-thought prompting. Further investigation into alternative forms of chain-of-thought prompting will hopefully inspire future advancements in language-based reasoning.